

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Python for Data Science Practical

Practical - V

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Class : SY IT	Batch :
Date of Assignment :	Date/Time of Submission : 12-8-2026

Part – A Theory

5a. Write a python code to demonstrate importing pandas libraries and create data frame object.

code:

```
import pandas as pd

data = {
    "name": ["sajid", "khalid", "farhan"],
    "age": [19, 22, 50],
    "subject": ["BSCIT", "BCOM", "BCA"],
    "Score": [60, 70, 80]
}

df = pd.DataFrame(data)
newdf = df.filter(items=["name", "age"])
print(newdf)
```

Output:

```
=====
   name  age
0  sajid   19
1  khalid   22
2  farhan   50
```

5b. Write a python code to show statistical information on given data set.

code:

```
import statistics as st

print(st.harmonic_mean([40, 60, 70]))
print("harmonic mean")
print(st.mean([1, 2, 3, 4, 5, 6, 11]))
print("mean")
print(st.variance([1, 2, 5, 6, 11]))
print("variance")
print(st.stdev([1, 2, 3, 70]))
print("stdev")
print(st.pstdev([4, 5, 6, 11]))
print("pstdev")
```

output:

```
53.61702127659574
harmonic mean
4.571428571428571
mean
15.5
variance
34.009802508492555
stdev
2.692582403567252
pstdev
```

5c. Write a python code to create pandas series from dictionaries.

Code:

```
import pandas as pd

dictionary = {'name': "sajid", 'course': "BSCIT", 'score': 60}
series = pd.Series(dictionary)

print(series)
```

Output:

```
-----
name      sajid
course    BSCIT
score      60
dtype: object
```

5d Write a python code to demonstrate filter pandas series with Boolean arrays.

Code:

```
import pandas as pd
dict = {
    'name': ["sajid", "khalid", "farhan"],
    'age': [19, 22, 50],
    'course': ["BSCIT", "BCOM", "BCA"],
    'Score': [60, 70, 80]
}
df = pd.DataFrame(dict, index = ["SY", "FY", "TY"])
print(df)
```

output:

```
----- RESIAMI. C. /
      name  age  course  Score
SY  sajid   19   BSCIT     60
FY  khalid   22   BCOM     70
TY  farhan   50    BCA     80
```

PART – C Curiosity Question:

